Adaptive Speaking Style Did Not Hold Up

Geist Atlas · Module 13 · What the Tutor Needs to Know

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[**SCOPE-BOUND**] Supported within a narrow scope · **Family:** What the Tutor Needs to Know
Claim: A pre-registered two-family test did not find a reliable benefit from adapting speaking style to the learner profile. Terra failed the profile-manipulation gate; Sonnet passed that gate but returned no interaction.

A 120-dialogue confirmatory study in one simulated detective world. It closes this field selector on the present evidence, while keeping the result separate from any claim about an isolated tutor model or human learning.

Derived view of `paper-full-2.0.md` v3.0.228 — §6.17. The paper is canonical; edit it there, not here.

6.17 Register Selection Does Not Produce a Confirmed Profile-Contingent Benefit Across Two Model Families (Pre-Registered Gate)

The register-policy line asked whether a tutor should change its speaking stance according to a learner profile, rather than whether one register is best in aggregate. A frozen confirmatory design crossed three policies (`bland`, the adaptive `field` selector, and a fixed `negative` stress register) with four v3 simulated-learner contracts (`diligent`, `affective_resistant`, `false_memory`, `proof_skipper`) at $n = 5$ per cell on `world_005_marrick`, once on an all-Terra stack and once on an all-Sonnet-5 stack: 60 dialogues per family, 120 total. The primary endpoint was mechanically computed proof-DAG coverage at learner turn 16. Each family also had to pass its own in-run profile-discrimination gate — average pairwise cosine < 0.85 and maximum similarity to `diligent` < 0.90 — before any policy-by-profile interaction could

*This sentence is the only one actually authored by Liam Magee in this paper.

confirm. A block that failed this manipulation gate was instrument-invalid for the family claim, regardless of an attractive cell contrast.

Whole stack	Selected traces	Average profile cosine	Maximum similarity to diligent	Frozen manipulation gate	Interaction verdict
Terra (`gpt-5.6-terra` at all four seams)	60	0.812	0.912	**fail**	instrument-invalid
Sonnet 5 (Sonnet at all four seams)	60	0.645	0.694	**pass**	null

Strict result: no family confirmation; no two-family claim. Terra's only supported policy-by-profile contrast was produced by the fixed hostile floor: the `negative` contrast for `affective_resistant` relative to its diligent contrast was -0.334 (95% bootstrap interval $[-0.468, -0.167]$). It ran opposite to the pre-registered beneficial-crossing direction and the family failed the binding manipulation gate because its maximum similarity to diligent was 0.912. It is therefore descriptive evidence of a possible stack-bounded contraindication, not a register-selection positive. Sonnet passed the frozen cosine gate, but every primary interaction interval crossed zero. The adaptive `field` policy confirmed no interaction on either stack. Excluding the four Sonnet rows whose runs ended after their valid turn-16 assessment but before full secondary closure leaves 56 rows and the same null verdict.

The comparison is deliberately read as a whole-stack boundary, not a tutor-model isolation. Tutor, automated learner, classifier, and learner-record extractor changed together between Terra and Sonnet; the coverage aggregate is deterministic, but the record from which it is computed still passes through an LLM extraction seam. The large cross-stack level difference can therefore arise from tutor conduct, learner conduct, extraction calibration, or their interaction. What closes is the general claim that this adaptive selector produces a profile-contingent fixed-horizon benefit on this apparatus. The register palette remains a useful implementation vocabulary, and Terra's hostile-register collapse remains a safety hypothesis, but neither licenses another selector-rescue block.

Status and caveats (per §5.12.6). Pre-registered and complete at $n = 5$ per cell; one detective world, simulated contract learners, one operational fixed-horizon endpoint, and no human-learning claim. Terra has complete secondary traces for all 60 selected rows; Sonnet has 60 primary-complete rows, 56 with full secondary closure, and four primary-only rows. One instrument boundary discovered after the fact: the Sonnet family block ran before the safe-mode-v1 claude-CLI context-isolation fix, so its speaking-tutor calls ambiently loaded ~16k tokens of repository context (the codex/Terra path was always isolated). This is a whole-family transport condition, uniform within the block — it cannot manufacture the within-family register contrasts, but it adds one

more uncontrolled difference to the already whole-stack Terra-vs-Sonnet comparison above, and Sonnet rows from this block must not be pooled with post-fix runs (check run-header provenance before any cross-run pooling; the §6.18-addendum Step 4 run is entirely post-fix). Reproduction is zero-call and hash-verified: the explicit final-row selection, 5,000-draw bootstrap (seed 20260713), profile gates, QA matrix, archive lineage, and strict verdict live in `scripts/analyze-register-confirmatory-step2.js`, `exports/register-confirmatory-evidence/final/`, and `config/adaptive-tutor-evidence/tutor-stub-reg`
The next experiment changes both delivery channel and intervention grain: point-of-action coaching and compiled constraints are a successor test, not a retry of the selector.

Neighbouring modules: Module 7 — A Bigger Learner Model Did Not Help; Module 3 — The Tutor Did Not Become More Adaptive; Module 12 — More Layers Do Not Mean More Benefit; Module 4 — How We Checked Every Claim.